

Linear Discriminant Analysis (LDA)

Introduction

Linear Discriminant Analysis (LDA) is a dimensionality reduction technique that is used in classification problems in which the data is assumed to have an underlying Gaussian multivariate distribution, with the same covariance matrix for each class. With LDA, the main goal is to project the data into a smaller-dimension subspace in which class separability is maximized.

Mathematical intuition

We want to find a projection vector, W , that maximizes the Fisher's linear discriminant

$$J(W) = \frac{|W^T S_B W|}{|W^T S_W W|}$$

where S_B is the between-class scatter matrix, and S_W is the within-class scatter matrix.

Both matrices are defined as

$$S_B = \sum_{k=1}^K n_k (\mu_k - \mu)(\mu_k - \mu)^T$$

$$S_W = \sum_{k=1}^K \sum_{x \in C_k} (x - \mu_k)(x - \mu_k)^T$$

Where μ is the global average, μ_k is the class k average.

In order to maximize Fisher's linear discriminant, we can differentiate the function and set it equal to zero. Using elemental identities from matrix calculus, it can be rewritten as

$$S_W^{-1} S_B W = \lambda W$$

with $\lambda = J(W^*)$. Therefore, we have an eigenvalue problem, in which the eigenvector corresponding to the highest eigenvalue is chosen as the projection.

What's linear about LDA?

From Bayes' rule, we can write

$$P(Y = k|X = x) = \frac{P(X = x|Y = k) \cdot P(Y = k)}{\sum_{i=1}^K P(X = x|Y = i)P(Y = i)}$$

Since we assumed data to follow a multivariate normal distribution with the same covariance matrix for each class, it can be shown that when we set a decision boundary

$$P(Y = k|X = x) = P(Y = j|X = x)$$

We end up with an expression for it that is linear with x . In contrast with QDA, for example, in which there is a different covariance matrix for each class, it is not possible to write such boundary in linear form.

Results

With this model, data is projected into a lower-dimensional subspace, which is helpful for classification problems without very complex or nonlinear relationships.

